

POCKET SMART AI — Your Smart Budget & Recommendation Assistant

Phase 6: Project Testing

St. George's Arts and Science College (UNM1727) · TN Skills | Generative AI · Team ID: SWTID-2026-2087

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6.1 Purpose of This Phase

This phase validates that Pocket Smart AI meets its functional and non-functional requirements through unit testing, integration testing, and user acceptance testing, and defines the performance metrics to be captured on the final build.

6.2 Unit Testing

Individual validation, calculation, allocation, recommendation, and formatting functions are tested independently — for example, confirming that a total-budget check correctly rejects an over-allocation before it ever reaches the recommendation layer.

6.3 Integration Testing

Verifies that browser input reaches the correct application logic and that processed results are returned and displayed correctly end-to-end — from the planner form, through backend orchestration and the AI recommendation call, to the rendered results page.

6.4 User Acceptance Testing (UAT)

Representative users complete the full planning flow (register/login → choose a planner → submit a budget → review results) to confirm that the screens, terminology, calculations, and recommendations are understandable to a non-technical audience.

6.5 Performance Metrics

For a web application, the team records response time, page-load behavior, concurrent-load observations, and security-validation results. Machine-learning-style metrics (accuracy/precision/recall/F1) are recorded only where genuinely meaningful and measured — this report avoids fabricating figures that were not supplied.

6.6 Representative Test Case Matrix

ID	Input	Expected Output	Status
TC-01	Open application	Landing/access screen loads	To be executed
TC-02	Valid registration data	Account registration completes	To be executed
TC-03	Invalid/empty required field	Validation message shown	To be executed
TC-04	Valid login credentials	User enters application	To be executed
TC-05	Budget + valid requirements	Allocation/recommendation generated	To be executed
TC-06	Budget exceeded by inputs	System prevents invalid total or explains constraint	To be executed
TC-07	Invalid numeric input	Input rejected safely	To be executed
TC-08	Result/dashboard request	Processed result rendered	To be executed
TC-09	Unexpected/service failure	Friendly error/fallback shown	To be executed

ID	Input	Expected Output	Status
TC-10	Production URL	Application accessible over intended deployment URL	To be executed

6.7 Testing Scope Notes

- Status values are marked 'To be executed' pending a full run against the final submitted build; results should be updated once the suite has actually been run.
- Security-validation checks (input sanitation, secret exposure, auth handling) should accompany functional test runs.
- Cross-planner coverage should include Home Interior, Party, and Jewelry flows, since each has distinct input fields.

6.8 Outcome of This Phase

A defined testing strategy spanning unit, integration, and acceptance levels, plus a 10-case representative test matrix that gives the team a repeatable checklist to execute and report against before deployment sign-off.